

Reference excerpt 01-020

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$z = z_{RA}$) the pressure feels the effects of the RA activity varying periodically with the cardiac cycle. The pressure at $z = z_{RA}$ (see Fig. 1) is supposed to be the sum of a component p_c constant in time and of a periodic component $p_a(t)$ which varies according to the atrial cardiac activity. The blood flow is due to the pressure gradient between the ends of the cylinder. It produces a velocity $w_T(t)$ that results from a constant component w_0 due to the constant gradient $p_c - p_r$ IG and a time varying component $w(t)$ due to the time varying $p_a(t) - p_r$ IG. This system can be described using the Womersley equation that is a linear differential equation where the unknown is the function w and the pressure gradient ($p(t)/z$) is the source term:

$$2w(r, z, t) r^2 + 1 r w(r, z, t) r - w(r, z, t) t = 1 p(t, z) z \quad (1)$$

The radial component of the blood velocity and the non-linear terms of the equation are negligible for $c \gg w$. [Womersley(1955c)] The solution of Eq.1 and the complete procedure to calculate it are explained in detail in [Sisini et al.(2015b)]. However the solution is there defined up to the multiplicative constant c and up to the additive constant w_0 . As a consequence, the instantaneous velocity trace, calculated in this way, is proportional but not equal to the actual blood velocity. Pressure waves propagation The pressure is transmitted along the direction AR-IJV according to the following equation